

12(a)	• emission from radioactive daughter products • self-absorption in source • absorption in air before reaching detector • detector not sensitive to all radiations • window of detector may absorb some radiation • dead-time of counter • background radiation <i>Any two points.</i>	B2
12(b)(i)	curve is not smooth or curve fluctuates/curve is jagged	B1
12(b)(ii)	clear evidence of allowance for background	B1
	half-life determined at least twice	B1
	half-life = 1.5 hours <i>(1 mark if in range 1.7–2.0; 2 marks if in range 1.4–1.6)</i>	A2
12(c)	1. half-life: no change	M1
	because decay is spontaneous/independent of environment	A1
	2. count rate (likely to be or could be) different/is random/cannot be predicted	B1